CONTRIBUTION RECORD - CSA06 CO3 Assignment

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The contribution record reflects the actual work completed. Primary contribution does not imply exclusive responsibility. Both students remain accountable for the complete assignment.

Question	Student A - Primary Contribution	Student B - Primary Contribution	Joint Work
Q1	Max/Min decomposition drawing, combine step, derivation of 3n/2 - 2 and comparison with 2(n-1). Binary Search experiments: successful/unsuccessful cases, comparison counts, increasing sizes.	Max/Min experiments: datasets, comparison counting, predicted vs observed, explanation of discrepancy. Binary Search elimination logic, decision rule, termination and efficiency argument.	Understanding both problems, strategy, base cases, implementation and testing of both algorithms.
Q2	Quick Sort investigation: balanced and unbalanced inputs, partition sizes, maximum	Merge Sort decomposition analysis, comparison with Quick Sort recursion,	Merge/Quick Sort implementation, first-element pivot choice, experiments, joint design

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	recursion depth, effect of pivot selection.	predictability, tree height, how imbalance gives O(n^2).	question (what to know about the input).
Q3	Integer decomposition, subproblem formation, why three multiplications, recurrence T(n) = 3T(n/2) + cn, solution Theta(n^1.585).	Crossover experiments: input sizes, repeated timing, multiplication counts, crossover range, small-input overhead.	Standard and Karatsuba implementations, crossover experiment setup, joint conclusion on asymptotic vs practical performance.
Q4	Experiments: matrix sizes, multiplication counts, repeated timing, practical crossover, addition/recursion/padding overhead, memory and cache effects.	Algebraic design: block decomposition, 8 vs 7 multiplications, correctness proof, padding.	Study of block multiplication, Strassen reconstruction, implementation and verification, joint engineering decision.
Q5	D&C decomposition design: organising points, dividing, what recursive calls return, combine step, boundary strip.	Comparative evidence: equivalent datasets, distance counts, execution behaviour, growth comparison, practical crossover.	Review of CO2 brute force, D&C design, implementation, comparison; cross-explanation of strip and evidence.

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Q6	Median of Medians: implementation, grouping, pivot, worst-case guarantee, QuickSelect comparison, deterministic poor pivot, randomisation. Verification answer on Meet-in-the-Middle storage.	Meet-in-the-Middle: division, subset generation, combination, search space, memory, comparison with 2^n, time-memory trade-off. Verification answer on pivot quality.	Cross-understanding of both investigations.
Q7	Independent initial design A, mathematical validation (recurrence and solution), cross-defence on experimental evidence.	Independent initial design B, experimental validation (test cases, instrumentation, baselines), cross-defence on how the recurrence arises.	Design comparison, joint final design, pseudocode, joint evaluation and conclusion.

Q7 independent design proposals

Student A's and Student B's independent Stage-1 design proposals are stored in Q7_Day4_Design_Challenge/Student_A_Initial_Design and Student_B_Initial_Design; the design comparison and joint final design are in Joint_Final_Design.

Signature of Student A: shahid

Signature of Student B: AZEEJ

Date: **2-11-2026**